BCSC0006: DATA STRUCTURES AND ALGORITHMS

COURSE OBJECTIVE

The objective of this course is that students will construct and application of various data structures and abstract data types including lists, stacks, queues, trees and graphs

CREDITS: 4 **L-T-P-J:3-1-0-0**

Module No.	Content	Teaching Hours
I	**Introduction**: Basic Terminology, Elementary Data Organization, Properties of an Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic Notations – Big-Oh; Operations on Data Structure, Abstract Data Types (ADT). **Linked Lists**: Implementation of Singly Linked Lists, Doubly Linked List, Circular Linked List, Operations on a Linked List - Insertion, Deletion, Traversal; Generalized Linked List, Polynomial Representation and Addition. **Stacks**: Primitive Stack Operations - Push & Pop, Array and Linked Implementation of Stack, Application of Stack: Prefix and Postfix Expressions, Evaluation of Postfix Expression, conversion of Infix to Postfix expression **Recursion**: Principles of Recursion, Tail Recursion, Removal of Recursion, use of stack in Recursion, Tower of Hanoi Problem. **Queues**: Operations on Queue - Add, Delete operations, Implementation of Queue Using Array and Linked List, Circular Queues, Deque and Priority Queue.	20 hours
II	**Trees:** Basic Terminology, Array Representation and Dynamic Representation; Complete Binary Tree, Algebraic Expressions, Extended Binary Trees, Tree Traversal Algorithms - Inorder, Preorder and Postorder; Threaded Binary Trees, Traversing Threaded Binary Trees. **Search Trees:** Binary Search Trees (BST), Insertion and Deletion in BST, AVL Trees, Introduction to M-Way Search Trees, B Trees. **Searching:** Sequential Search, Binary Search. **Sorting:** Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Two Way Merge Sort, and Heap Sort. **Graphs:** Terminology, Adjacency Matrices, Adjacency List, Graph Traversal - Depth First Search and Breadth First Search; Spanning Trees, Minimum Cost Spanning Trees – Prim's and Kruskal's Algorithm; Shortest Path Algorithm – Bellman-Ford and Dijkstra's Algorithm. **Hashing & Indexing:** Hash Function, Collision Resolution Strategies. Primary Indices, Secondary Indices, Indexing and Hashing Comparisons.	**20** hours

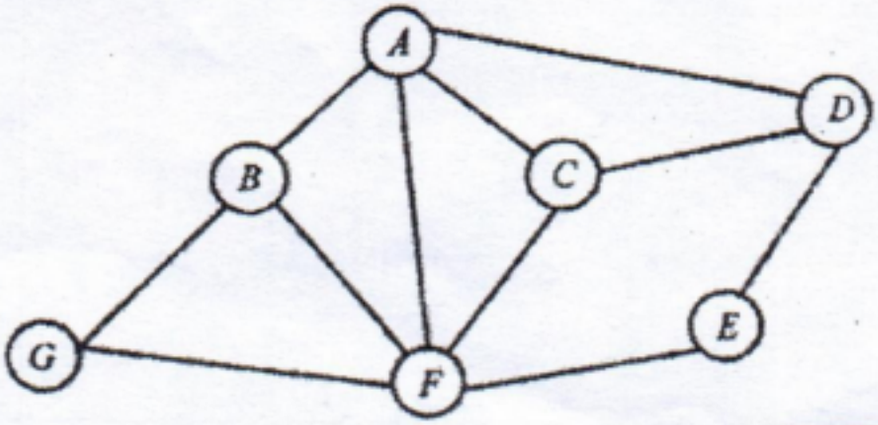
End Term Examination, Even Semester 2021-22
B.Tech (CSE/CCV/DA/IIOT/AIML/CSF), 2nd Year, 4th Semester
BCSC 0006 / Data Structures and Algorithms

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: Write the complete code in one place neatly. Commenting the code is optional.

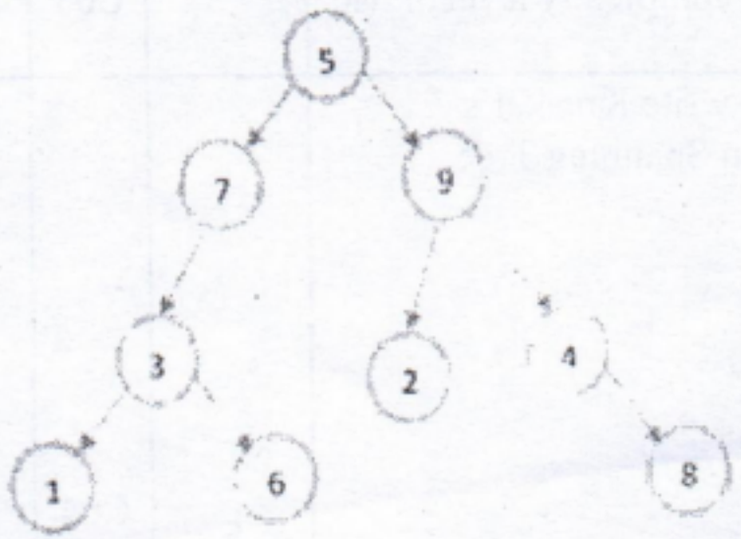
Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Assume that a Singly linked list is implemented with a header node but no tail node. Write a class that includes methods to a) Add a value X if it is not already contained in the linked list. b) Remove the value X if it is contained in the linked list.	2 2	C01 C02	C	C
2	What are the disadvantages of representing a linear queue using array? How are they overcome?	4	CO3	U	C
3	What is a Linked list? What will you prefer to use a singly or a doubly linked lists for traversing through a list of elements?	4	C01 C02	U, An	F
4	Represent following graph in adjacency list. Write the code for breadth first search to traverse the graph.  	4	C04 C05	U, A	C,F
5	Write an algorithm delete a node from AVL Tree.	4	C04	U, C	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

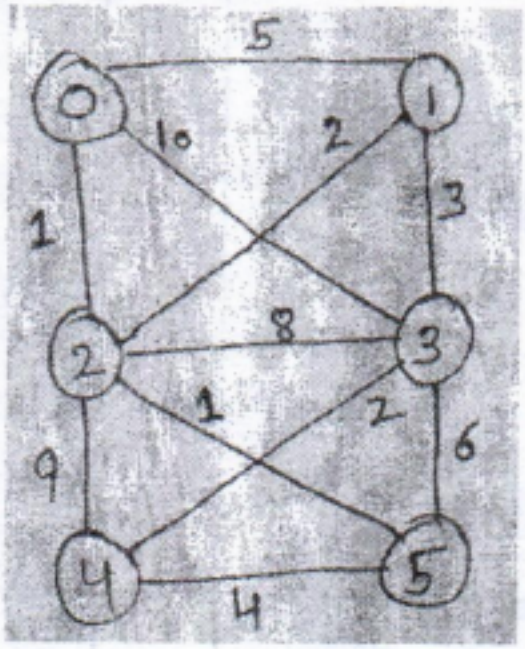
No.	Detail of Question	Marks	CO	BL	KL
1	Explain the various ways in which a graph can be represented bringing out the advantages and disadvantages of each representation.	1.5 1.5	C04	E C	C
2	a) What is the output obtained after preorder, postorder traversal of following tree.  b) Draw the binary tree whose sequential representation is given below.	3	C05	An	M,C

1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	-	E	F	-	G	-	H	I	-

3	Define Hashing. What is the need for Hashing? Define collision problem in Hashing?	3	C05	C	M,C
4	What is a binary search tree (BST)? Write the code to construct a BST with five nodes.	3	C05 C04	C,A	C
5	Write Short Node on a) AVL Tree b) Threaded Binary Tree c) Minimum Spanning Tree	1 1 1	C05	R	C,F

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Write algorithm for (i) Insertion sort (ii) Bubble sort. Illustrate the insertion sort algorithm and bubble sort algorithm on input [30,20,10,60,80,40]	2.5 2.5	C05	A	CF
2	Why Binary Search algorithm is more efficient that linear search? Depict your answer with suitable example? Mention the time complexity level of two algorithms.	5	C04 C05	U,A	C
3	For the graph given below, write Kruskal's Algorithm to find Minimum Spanning Tree.  	5	C04 C05	A	CF

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End Term Examination, Even Semester 2018-19
B.Tech (CSE/CCV/DA), 2nd year, 4th Semester
BCSC0006 : Data Structure & Algorithms

Time: 3 Hour **Maximum Marks: 50**

Note: Write the complete code in one place neatly. Commenting the code is optional.

Section-A **7 x 5 = 35 Marks**

Q1. Define Data Structures. What is the difference between Linear and Non-linear data structures? What are the operations that can be performed on data structures.

Q2. Discuss the specification of a Binary Search Tree. Explain the different cases to delete an element from binary search tree.

Q3. What do you understand by stack. Explain the condition of Stack overflow and underflow.

Q4. Modify the body of the following method such that, if the input to the method is 33, 44, 55, 66 then, the method will return 4.

```
int size(Node list) {
// write your code here
}
```

Q5. Write program to implement doubly linked list provide following operations.

I. Insert at beginning
II. Insert at location
III. Remove from beginning
IV. Remove from location

Q6. Give ADT for Queue Data Structure. Discuss in brief any two applications of queue data structure.

Q7. Write code to Construct Binary search Tree from the following elements:-
12,8,25,14,9,6,18. Also show preorder, inorder and postorder traversal for the same.

Section-B

3 x 2 = 6 Marks

Q1. Sort the following data using merge sort and selection sort.

142,317,45,222,187

Q2. Define Sorting. What is the importance of Sorting?

Q3. Explain the term recursion with an example.

3 x 3 = 9 Marks

Q1. Explain with suitable example, BFS and DFS traversal of a graph.

Q2. What is AVL tree? Explain with suitable example the RR rotation and balance Factor.

Q3. Define the following Terms with respect to Trees:

A. Root
B. Subtree
C. Level of Node
D. Depth of Tree
E. Sibling
F. Forest

End Term Examination, Odd Semester 2017-18
Course: B.Tech. (CSE), Year: II
Data Structures & Algorithms (CSE-3001)

Uni. Roll No:
Time: 02 ½ Hours **Total Marks: 40**
Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. Compare Iteration with Recursion.

II. Which is the best data structure to check an expression has balanced parentheses?

III. Explain Non-Linear Data Structures.

IV. Write the polish notation of (A - B) / (C * D) - E

V. Write the applications of Queue in Computer Science.

VI. Write the condition of overflow and underflow in Circular Queue.

VII. Construct the expression tree for the given expression

(A *B - C) + ((D + E * F) + G)

VIII. Write the properties of B-Tree.

IX. What is the minimum number of nodes in a AVL tree of height 4?

X. How many distinct binary search trees can be created out of 4 distinct keys?

XI. A 3-ary tree is a tree in which every internal node has exactly 3 children. Find the number of leaf nodes in such a tree with 6 internal nodes.

XII. Write the complexity of Quick sort in average and worst case.

XIII. Compare BFS with Dijkstra's Algorithm.

XIV. Explain linear probing.

XV. Which data structures are used in Prim's and Dijkstra's Algorithm?

XVI. How records are inserted in sequential files?

Section B

Note: Attempt Any Four Questions **(3x4=12)**

I. Consider a hash table of size $m = 1000$ and a corresponding hash function $h(k) = \text{floor}(m\ (kA \bmod 1))$ for $A = (\sqrt{5} - 1)/2$. Compute the locations to which the keys 61, 62, 63, 64, and 65 are mapped.

II. Let P be a shortest path from some vertex s to some other vertex t in a graph. If the weight of each edge in the graph is increased by one, whether P will still be a shortest path from s to t. Explain with example.

III. Explain the recursive solution to all-pair shortest path problem.

IV. Give an adjacency-list representation for a complete binary tree of 7 vertices. Give an equivalent adjacency-matrix. Explain the disadvantages of adjacency list representation.

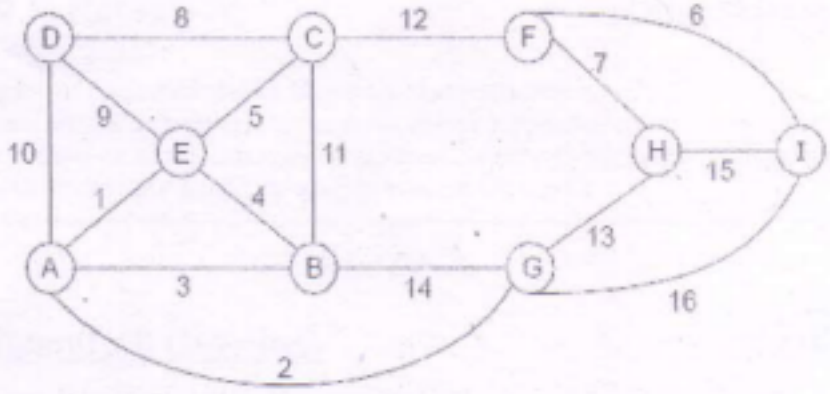
V. Compare Indexing with Hashing.

Section C

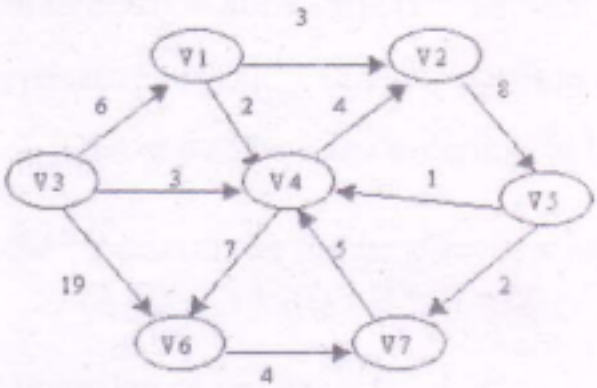
Note: Attempt Any three Questions **(4x3=12)**

I. Explain Sparse, Dense, Primary and Secondary indices with suitable example.

II. Explain two applications of minimum spanning trees in Computer Science. Apply Prim's Algorithm on the given graph. Consider E as source vertex.



III. Discuss the similarity between Dijkstra's Algorithm and Prim's Algorithm. Apply the Dijkstra's Algorithm on the given graph. Consider V3 as a source vertex.



IV. Discuss the problems involved in linear probing. Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length m=11 using open addressing with the auxiliary hash function $h_1(k) = k$. Illustrate the result of inserting these keys using quadratic probing with $c1 = 1$ and $c2 = 3$, $h_2(k) = 1 + (k \bmod (m-1))$.

Printed Pages: 02 **University Roll No.....**

End Term Examination, Odd Semester, 2016-17

CSE 3001 : Data Structures & Algorithms

Time: 02 ½ Hours **Max. Marks: 40**

Section – A

Note: Attempt All Questions. **(1x16=16)**

I. Define Abstract Data Type (ADT).
II. What are called Dynamic data structures?
III. What is worst-case efficiency?
IV. Minimum number of queues needed to implement the priority queue?
V. What is circular linked list?
VI. What is recursion?
VII. What is binary search?
VIII. Define the Complete Binary Tree.
IX. Write a difference between B trees and B+ trees.
X. In an AVL tree, at what condition the balancing is to be done?
XI. What is the difference between Inorder and Postorder tree traversal?
XII. What is the difference between binary tree and binary search tree?
XIII. What is the difference between a tree and a graph?
XIV. What is minimum spanning tree?
XV. What is Chained Hash Table?
XVI. Define the complete graph.

Section – B

Note: Attempt Any Four Questions. **(3x4=12)**

I. Explain Collision resolution strategies.

II. What is difference between file structure and storage structure?

III. What are the properties of spanning trees?

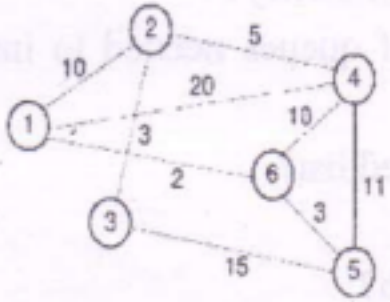
IV. What is Breadth First Searching in graphs? Which data structure is suitable for BFS algorithm? Justify your answer.

V. What are the methods available to storing sequential files?

Section – C

Note: Attempt Any Three Questions. **(4x3=12)**

I. Draw the adjacency matrix representation for the following graph.



II. Assume that you are hashing key K to a hash table of n slots (indexed from 0 to n − 1). For each of the following functions h(K), is the function acceptable as a hash function (i.e., would the hash program work correctly for both insertions and searches), and if so, is it a good hash function or not? Justify your answer (**Note: Function Random(n) returns a random integer between 0 and n − 1, inclusive**).

a) h(k) = k/n where k and n are integers.

b) h(k) = 1.

c) h(k) = (k + Random(n)) mod n.

d) h(k) = k mod n, where n is a prime number.

III. Write the algorithm for DFS.

IV. What are the different techniques used in allocating file blocks? Explain the pros and cons of each technique.

Printed Pages-3 University Roll No.....

End Term Examination, Odd Semester 2015-16

CSE3001: Data Structures & algorithms

Time: 02 ½ Hours Total Marks: 40

Instruction:

1. Attempt all questions from Section-A, any four from Section-B and any three from Section-C.
2. Symbols have their usual meanings.

Section A

Note: Attempt All Questions **(1x16=16)**

I. List out the areas in which data structures are applied extensively?

II. If the characters 'P', 'Q', 'R', 'S' are placed in a stack (in that order), and then removed one at a time, in what order will they be removed?

III. How many null nodes will a binary tree contains with 20 nodes.

IV. There are 8, 15, 13, 14 nodes were there in 4 different trees. Which of them could have formed a full binary tree?

V. Write down the two applications in computer science of stack data structure.

VI. In an AVL tree, at what condition the balancing is to be done?

VII. Draw all possible nonsimilar trees where T is a binary tree with 3 nodes.

VIII. What is the complexity to search an element in worst case in BST.

IX. Write down the application areas where linked list is useful.

X. Does the minimum spanning tree of a graph give the shortest distance between any 2 specified nodes?

XI. Whether Linked List is linear or Non-linear data structure?

XII. What is the complexity of quick sort algorithm?

XIII. In a selection sort of n elements, how many times is the swap function called in the complete execution of the algorithm?

XIV. What is hashing?

XV. Write down the name of data structures used in DFS and BFS.

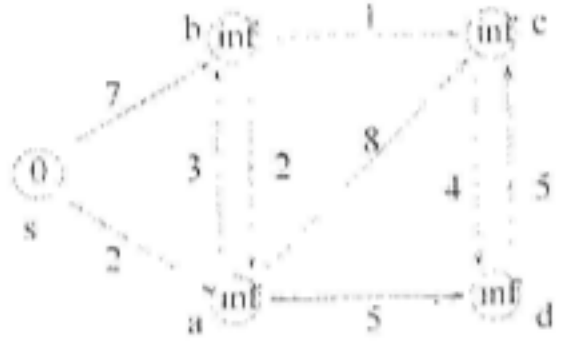
XVI. What should be the depth of a complete binary tree with n elements?

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Explain the structure operations and disadvantages of sequential files.

II. Write down the steps of Dijkstra' algorithms and implement on the following graph



III. Enlist the basic differences between tree and graph. Discuss different technique for storing graph.

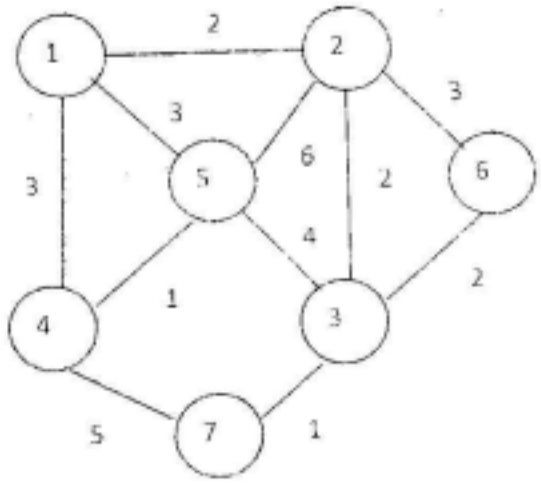
IV. What is linear probing? How it differ from quadratic probing?

V. What is Breadth first Search traversal? Explain its algorithm with an example.

Section C

Note: Attempt any Three Questions **(4x3=12)**

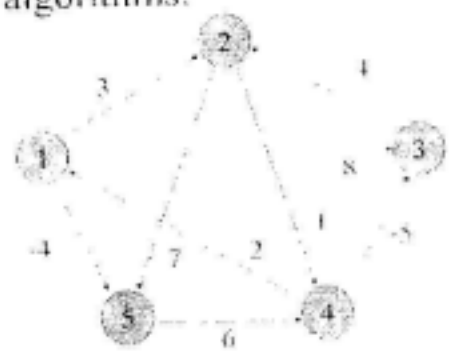
I. What is spanning tree? When it can be called a minimum spanning tree? Find Minimum spanning tree for the following Graph (G) by using Kruskal's Algorithm.



II. Using a hash table with eleven slots indexed through 0 to 10 and the hashing function h(k)= k mod 11, show the hash table that results when the following keys are inserted in the order given:
26, 42, 5, 44, 92, 59, 40, 36, 12, 60, 80.
Assume that collisions are resolved using synonyms chaining.

III. How insertions and deletions of records are handled in indexed sequential files.

IV. Compute the shortest- path between using Floyd-Warshall algorithms.



Printed Pages: 02 **University Roll No.....**

End Term Examination, 2014-15

B. Tech. Year: II, Semester: III

CSE 211: Data Structures and Algorithms

Time: 2 ½ Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemente with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. What is the advantage of using a Circular Queue over a queu

II. Draw expression tree for the expression (a + ((b*(c-e))/f)..

III. Write the recursive function for preorder traversal of BST

IV. Write the difference between cursor and th

V. Explain theta Notation in detail.

VI. If the number of leaf nodes in a strictly
the number of edges in the tree.

VII. Define MST with an example.

VIII. Define Quadratic Probing

IX. Write complexity of quicksort algorithm for all the three cases.

X. Instead of using counting sort to sort digits in the radix sort algorithm, we can use any valid sorting algorithm and radix sort will still sort correctly. Give your answer(T/F) and also justify it.

XI. Give four applications of stack.

XII. Define node, degree, siblings, depth/height, level

XIII. How do we store a graph in Computer's memory? Discuss.

XIV. Define M-way Search Tree.

XV. Give four example of in-place sorting algorithm.

XVI. Compare sequential search with Binary Search.

Section B

Note: Attempt any FOUR Questions **(3x4=12)**

I. Illustrate the operation of RADIX-SORT on the following list of English words:

COW, DOG, SEA, ROW, MOB, BOX, TAB, BAR, EAR, BIG, TEA, NOW

II. Write any sorting algorithm based on divide and conquer approach

III. Write a recursive algorithm to search an element in an array using Binary Search.

IV. Write an algorithm for Insertion Sort. Show steps(passwise) to sort the given array using Insertion Sort: **3, 5, 43, 2, 0, 1, 32, 22, 14, 34**

V. Write an algorithm for Selection Sort. Show steps(passwise) to sort the given array using Selection Sort: **19,23,14,9,5,2,7,4,1**

Section C

Note: Attempt any THREE Questions **(4x3=12)**

I. Write a recursive algorithm for Merge Sort. Apply Merge sort algorithm for sorting following integer array. **1, 12, 8, 2, 25, 6, 14, 28, 17, 7, 52, 16, 48, 68, 3, 2**

II. Given input **{ 4371,1323,6173,4199,4344,9679,1989}** .Find the h(X) by applying (m=10)

c) Division Method

d) Mid square Method

III. Write algorithm of Heapsort. Sort the following elements using Heap Sort Algorithm.(show all the steps):
6, 5, 3, 1, 8, 7, 2, 4

IV. Write short notes on File Structure.

Uni. Roll_____

GLA University, Mathura

Course: -B. Tech. II-Year, End Term (Odd Sem.) Examination, 2013-14

Subject: Data Structures and Algorithms (CSE-211)

Time: - 150 Minutes **M.M:-40**

Notes:-

1. Answer **ALL** questions from Section A, **ANY FOUR** from Section B and **ANY FOUR** from Section C.
2. All questions of the particular section should be answered collectively at one place.
3. Answer should be to-the-point and wherever required, be supplemented with neat sketches.
4. Any missing data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate marks.

Section-A (Answer all questions) 08 x 1 = 08

1. List out at least two applications of tree data-structure in Computer field?
2. There are 8, 15, 13, 14 nodes were there in 4 different trees. Which of them could have formed a full binary tree?
3. In RDBMS, what is the efficient data structure used in the internal storage representation?
4. The postfix form of the expression (A + B)*(C*D – E)*F / G is_____
5. Merging 4 sorted files containing 50, 10, 25 and 15 records will take_____time.
6. Suppose that an open-address hash table has a capacity of 811 and it contains 81 elements. What is the table's load factor? (An approximation is fine.)
7. Here is an array with exactly 15 elements:

   1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

   Suppose that we are doing a binary search for an element. Circle elements that will be found by examining two or fewer numbers from the array.
8. In the last node of the circularly doubly linked list, the link field contains_____.

Section-B (Attempt any **four** questions) 04 x 03 = 12

1. What is an index? What are the various types of indexing?

2. Draw an AVL tree mentioning the rotations at each insertion for the following set of elements and in the order given.

   JAN, DEC, FEB, MAR, APR, JUNE, JULY, NOV, OCT, SEPT, AUG, MAY

3. Write the functions to perform deletion and insertion in a DEQUE implemented using linked list.

4. Write an algorithm to implement two stacks using an array. Provision should be made such that any of these stacks can grow up to the size of the array if the other stack is not occupying the complete space left.

5. Construct a B+ tree of order 5 for the given set of elements –

90, 30, 120, 140, 25, 35, 160, 180, 10, 20, 50, 60, 40, 80, 100, 70, 130

6. Does *quicksort* takes $O(n^2)$ time when the input list is already in sorted order? Justify your answer.

Section-C (Attempt any **four** questions) 04 x 5 = 20

1.
```
void heapsort(element a[ ], int n)
{
int i, j;
element temp;
for (i=n/2; i>0;i--)
adjust_max_heap(a,i,n);
for(i=n-1;i>0;i--)
{
swap(a[1],a[i+1],temp);
adjust_max_heap(a,1,i);
}
}
```
Write the status of the list (12, 2, 16, 30, 8, 28, 4, 10, 20, 6, 18) at the end of the first ***for*** loop as well as at the end of each iteration of the second ***for*** loop of the given above ***heapsort***.

2. Write the status of the list (12, 2, 16, 30, 8, 28, 4, 10, 20, 6, 18) at the end of each pass of *radixsort*. Use 10 as the radix.

3. Write and explain the binary search algorithm with its complexities.

4. Discuss different types of collision resolution technique with suitable diagram.

5. Mention the *mergesort* algorithm. Explain it working with an example with at least 10 elements.

6. Mention the *bubblesort* algorithm which takes $O(n^2)$ in best case. Can this algorithm be modified such that the best case complexity will be O(n)? Justify your answer.

SET B

GLA University, Mathura

End Term Examination, 2012 – 2013

Course: - B. Tech. (CS) II Year, III Semester

Subject: Data Structures and Algorithms (CSE211) **University Roll No.**

Time: 2 Hour 30 Minutes **Total Marks: 40**

Notes:

1. **Part A is compulsory.** Answer **Any four** from **Part B** and **Any four** from **part C.**
2. All parts of a question should be answered at one place.
3. Answer should be brief-to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably with justification.
5. Figures on the right-hand side margin indicate full marks.

Part A

Attempt all questions. **[0.5 x 20 = 10]**

1. Which of the following data structure can't store the non-homogeneous data elements?
   - Pointers
   - Records
   - Arrays
   - None of these
2. Which of the following statement is false?
   - Arrays are dense lists and static data structure
   - Pointers store the next data element of a list
   - Data elements in linked list need not be stored in adjacent space in memory
   - Linked lists are collection of the nodes that contain information part and next pointer
3. A variable P is called pointer if
   - P contains the address of an element in DATA
   - P can store only memory addresses
   - P contain the DATA and the address of DATA
   - P points to the address of first element in DATA
4. Which of the following is not the required condition for binary search algorithm?
   - There should be the direct access to the middle element in any sub list
   - The list must be sorted
   - There must be mechanism to delete and/or insert elements in list
   - None of these
5. What is the time complexity of Heap sort?
   - O(n)
   - $O(n\log_2 n)$
   - $O(\log_2 n)$
   - $O(n^2)$
6. Binary search algorithm cannot be applied to
   - Sorted linked list
   - Sorted binary trees
   - Sorted linear array
   - Pointer array

7. In static hashing
   - The hash table size does not increase in any condition
   - The hash table size increases on collision
   - Both can be possible
   - Cannot predict
8. The memory address of the first element of an array is called
   - Floor address
   - Foundation address
   - First address
   - Base address
9. Which of the following algorithm is used for determining single-source shortest path in a weighted digraph (where some of the edge weights may be negative)
   - Dijkstra algorithm
   - Floyd-Warshall algorithm
   - Bellman-Ford algorithm
   - None of these
10. Identify the data structure which allows deletions at both ends of the list but insertion at only one end
    - Output-restricted deque
    - Priority queues
    - Input-restricted deque
    - None of these
11. A binary tree whose every node has either zero or two children is called
    - Complete binary tree
    - Binary search tree
    - Extended binary tree
    - None of these
12. The depth of a complete binary tree is given by
    - $D_n = n\log_2 n+1$
    - $D_n = \log_2 n$
    - $D_n = n\log_2 n$
    - $D_n = \log_2 n+1$
13. When converting binary tree into extended binary tree, all the original nodes in binary tree are
    - External nodes on extended tree
    - Vanished on extended tree
    - Internal nodes on extended tree
    - None of these
14. The post order traversal of a binary tree is DEBFCA. Find out the pre order traversal
    - ADBFEC
    - ABDECF
    - ABFCDE
    - ABDCEF
15. The in order traversal of tree will yield a sorted listing of elements of tree in
    - Binary search trees
    - Heaps
    - Binary trees
    - None of these
16. Reorder the sequence 4 15 8 18 9 11 10 so that when elements are inserted in this new order into an empty binary search tree, the resulting tree has height 3. Recall that by "height" I mean the number of nodes in the longest path from root to leaf.
    - 4 9 8 10 15 11 18
    - 10 8 15 9 4 11 18
    - 10 8 15 4 9 11 18
    - 10 8 4 15 9 11 18
17. The worst case time complexity of Quick sort is:
    - $O(n\log_2 n)$
    - $O(\log_2 n)$
    - $O(n)$
    - $O(n^2)$
18. Assume that a data fie has an index consisting of N items, where N is large. If a binary search of the index is used to find an item, then, of the following, which best approximates the mean number of comparisons required to locate a specified index entry?
    - $N(N+1)/2$
    - $\log_2 N - 1$
    - $(N+1)/2$
    - $N \log_2 N$

19. A queue data structure observes first in first out flow
    - The statement is true if queue is implemented with array but not when queue is implemented as linked list
    - The statement is true for non circular queues only
    - The statement is false as queue observes last in first out flow
    - The statement is true in all conditions
20. A tree is an acyclic graph
    - The statement is incorrect
    - The statement is correct
    - One cannot define one data structure in terms of other data structure so both the options are meaningless
    - None of these

Part B

Attempt any four questions. **[2.5 x 4 = 10]**

1. Given pointers to two nodes in a doubly linked list, write pseudo code for swapping the order of the elements.
2. Show with examples the insertion and deletion of elements in an array based and linked list based stack. What are key differences in both of these implementations?
3. Show the AVL tree inserting keys 3, 9, 5, 6, 1, 2, 11 and 10 in the given order.
4. Show the selection sort steps for given elements 13, 21, 53, 34, 93, 12, 121 and 100 to be sorted in ascending order.
5. What is the minimum height and maximum height of a binary tree with 32 elements? Why?

Part C

Attempt any four questions. **[5 x 4 = 20]**

1. Write algorithm for Merge sort or Quick sort. Do both of these have similar time complexity in Big O notation? Explain your answer.
2. Show with examples the insertion of 10 elements of your choice and then deletion of 4 elements in a B+ tree. Explain the terms overflow and underflow in the context of B+ tree.
3. Write a details note on storage media file organization.
4. Show the Min heap with keys 3, 9, 5, 6, 1, 2, 11 and 10 being inserted in the given order.
5. Show the bubble sort steps for given elements 33, 29, 55, 46, 11, 72, 141 and 10 to be sorted in ascending order.

GLA University, Mathura

Final Term Examination, 2011-12

Course: - B.Tech. II-Year, III-Sem.

Subject: - Data Structure And Algorithm **Uni. Roll No:-**

Time:-2 Hrs 30 Min. **Total Marks:-80**

Notes:-

1) Answer all questions, Group A is **compulsory, Any Four** from Group B and **Any three** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

GROUP/SECTION-A(1x20)

Select the correct answer:-

Q.1 Let A be an adjacency matrix of a graph G. The th ij entry in the matrix K A , gives
(A) The number of paths of length K from vertex Vi to vertex Vj.
(B) Shortest path of K edges from vertex Vi to vertex Vj.
(C) Length of a Eulerian path from vertex Vi to vertex Vj.
(D) Length of a Hamiltonian cycle from vertex Vi to vertex Vj.

Q.2 The OS of a computer may periodically collect all the free memory space to form contiguous block of free space. This is called
(A) Concatenation **(B)** Garbage collection
(C) Collision **(D)** Dynamic Memory Allocation

Q.3 You have to sort a list L consisting of a sorted list followed by a few "random" elements.Which of the following sorting methods would be especially suitable for such a task?
(A) Bubble sort **(B)** Selection sort
(C) Quick sort **(D)** Insertion sort

Q.4 B Trees are generally
(A) very deep and narrow **(B)** very wide and shallow
(C) very deep and very wide **(D)** cannot say

Q.5 A technique for direct search is
(A) Binary Search **(B)** Linear Search
(C) Tree Search **(D)** Hashing

Q.6 If a node having two children is deleted from a binary tree, it is replaced by its
(A) Inorder predecessor **(B)** Inorder successor
(C) Preorder predecessor **(D)** None of the above

Q.7 The number of interchanges required to sort 5, 1, 6, 2 4 in ascending order using Bubble Sort is
(A) 6 **(B)** 5
(C) 7 **(D)** 8

Q.8 The postfix form of the expression (A+ B)*(C*D– E)*F / G is

(A) AB+ CD*E – FG /** **(B)** AB + CD* E – F **G /

(C) AB + CD* E – *F *G / **(D)** AB + CDE * – * F *G /

Q.9 The complexity of multiplying two matrices of order m*n and n*p is

(A) mnp **(B)** mp **(C)** mn **(D)** np

Q.10 In worst case Quick Sort has order

(A) O (n log n) **(B)** O (n2/2) **(C)** O (log n) **(D)** O (n2/4)

Q.11 A full binary tree with 2n+1 nodes contain

(A) n leaf nodes **(B)** n non-leaf nodes

(C) n-1 leaf nodes **(D)** n-1 non-leaf nodes

Q.12 If a node in a BST has two children, then its inorder predecessor has

(A) no left child **(B)** no right child **(C)** two children **(D)** no child

Q.13 A binary tree in which if all its levels except possibly the last, have the maximum number of nodes and all the nodes at the last level appear as far left as possible, is known as

(A) full binary tree. **(B)** AVL tree. **(C)** threaded tree. **(D)** complete binary tree.

Q.14 A linear list of elements in which deletion can be done from one end (front) and insertion can take place only at the other end (rear) is known as a

(A) queue. **(B)** stack. **(C)** tree. **(D)** linked list.

Q.15 An undirected graph G with n vertices and e edges is represented by adjacency list. What is the time required to generate all the connected components?

(A) O (n) **(B)** O (e) **(C)** O (e+n) **(D)** O (2) e

Q.16 Consider a linked list of n elements. What is the time taken to insert an element after an element pointed by some pointer?

(A) O (1) **(B)** O (log2 n) **(C)** O (n) **(D)** O (n log2 n)

Q.17 The smallest element of an array's index is called its

(A) lower bound. **(B)** upper bound. **(C)** range. **(D)** extraction.

Q.18 In a circular linked list

(A) components are all linked together in some sequential manner.

(B) there is no beginning and no end.

(C) components are arranged hierarchically.

(D) forward and backward traversal within the list is permitted.

Q.19 One of the major drawback of B-Tree is the difficulty of traversing the keys sequentially.

(A) True **(B)** False

Q.20 O(N) (linear time) is better than O(1) constant time.

(A) True **(B)** False

Section B

Q1. a) A two dimensional array TABLE [6] [8] is stored in row major order with base address 351. What is the address of TABLE [3] [4]? (2)

b) What is an algorithm? What are the characteristics of a good algorithm? (3)

Q2: What are expression trees? Represent the following expression using a tree. Comment on the result that you get when this tree is traversed in Preorder, Inorder and postorder. (a-b) / ((c*d)+e) (5)

Q3: Show the result of inserting the keys. (5)

F, S, Q, K, C, L, H, T, V, W, M, R, N , P, A, B, X, Y, D, Z, E in the order to an empty B-tree of degree-3.

Q4: Which are the two standard ways of traversing a graph? Explain them with an example of each. (5)

Q5: Write an algorithm to create a single linked list. Write functions to do the following Operations (1+2+2)

(i) Insert a new node at the end

(ii) Delete the first node

Section C

Note: Question number 1is compulsory

Q1: a) The following values are to be stored in a hash table (7+7)

25, 42, 96, 101, 102, 162, 197

Describe how the values are hashed by using division method of hashing with a table size of 7. Use chaining as the method of collision resolution.

b) What do you mean by hashing? Explain any three popular hash functions.

Q2: a) What do you understand by file organization? List the various file operations. Explain them. (7+6)

b) Write binary search algorithm and trace to search element 91 in following list:
13 30 62 73 81 88 91
What are the limitations of Binary Search?

Q3: a) Write down the algorithm of quick sort. Also write its complexity for all the cases.

b) Sort the following array of elements using quick sort (show pass-wise). (7+6)

{3 1 4 1 5 9 2 6 5 3 5 8}

Q4: a) Write down the algorithm of Merge sort. Also write its complexity for all the cases. (6+7)

b) Sort the following sequence of keys using merge sort(show pass-wise).
66, 77, 11, 88, 99, 22, 33, 44, 55